Spotter Freight Rate Prediction Assessment

Machine Learning Model Development & Validation Report

Model: CatBoost Regression
Validation: Rolling Temporal Validation
Final Training Data: 48,000 labeled loads
Validation Predictions: 12,000 loads
December Scenario: 31 fixed-input predictions

October 2026

Spotter Freight Rate Prediction Assessment

1. Executive Summary

This project develops a machine learning model for freight load rate prediction using the provided Spotter development dataset.

The development dataset contains 48,000 labeled loads, while the final validation set contains 12,000 loads.

A temporal validation strategy was used to evaluate future prediction performance. The final model is a CatBoost regression model using route, equipment, distance, weight, and calendar features.

The final prediction files were successfully validated using the provided scoring script.

2. Data Overview

The development data contains 48,000 historical freight loads.

The validation set contains 12,000 loads.

The December scenario contains 31 rows from December 1 through December 31, 2025, with fixed route and load characteristics.

3. Data Cleaning

The weight feature contained missing values and negative values.

Negative weights were treated as sign errors and converted using:

weight = abs(weight)

Missing weight values were retained because CatBoost can handle missing numerical values.

4. Exploratory Analysis

Distance showed a strong relationship with posted rate, with a correlation of approximately 0.909 in the cleaned training data.

Equipment type also showed differences in average rate and rate-per-mile.

5. Baseline

The overall median rate-per-mile baseline achieved an October MAE of 253.85.

An equipment-specific median rate-per-mile baseline achieved an October MAE of 231.30.

6. Validation Strategy

A temporal validation strategy was used.

Primary split:

- Training: January 1, 2025 through September 30, 2025
- Validation: October 1, 2025 through October 31, 2025

Rolling temporal validation was also performed using July, August, September, and October as validation months.

7. Model

CatBoost regression was selected because it supports categorical features directly and handles missing numerical values.

Features:

- pickup
- delivery
- distance
- equipment
- weight
- month
- day_of_year
- day_of_week

Categorical features:

- pickup
- delivery
- equipment

Final configuration:

- iterations: 600
- learning_rate: 0.05

- depth: 7
- loss_function: MAE
- random_seed: 42

8. Validation Results

Validation Month	MAE
July	148.15
August	90.43
September	108.49
October	115.41
Mean	115.62

The October direct CatBoost model achieved an MAE of 115.41 during model selection.

9. Final Predictions

The final model was trained using all 48,000 cleaned training rows.

It generated:

- 12,000 validation predictions
- 31 December predictions

The validation file contains:

load_id,predicted_rate

The December file contains:

pickup,delivery,distance,equipment,weight,date,predicted_rate

10. December 2025 Scenario

Fixed inputs:

- Pickup: Lexington
- Delivery: Fort Wayne
- Distance: 360 miles

- Equipment: Dry Van
- Weight: 32,000 lb

Only the date changes across the 31 December rows.

The predicted rates are approximately between $831 and $851.

The provided scorer generated:

scorer_results/candidate_december.png

11. Submission Validation

The provided scoring script successfully validated:

- 12,000 final validation predictions
- 31 December predictions
- Validation IDs
- Positive prediction values
- December dates
- Fixed December inputs

The December chart was generated successfully.

Official final validation metrics are calculated by Spotter after submission.

12. Reproducibility

The project includes:

- requirements.txt
- README.md
- src/train_and_predict.py
- score.py
- validation_predictions.csv
- december_predictions.csv
- December prediction chart

13. Conclusion

The project provides a complete freight-rate prediction pipeline covering data cleaning, exploratory analysis, temporal validation, model development, final prediction generation, and scorer validation.

The selected CatBoost model improved over the equipment-specific rate-per-mile baseline on the October temporal validation split.

All required prediction files passed the provided scoring checks.

December 2025 Fixed-Input Scenario

The following chart shows the model predictions for the fixed December 2025 scenario specified by the assessment.

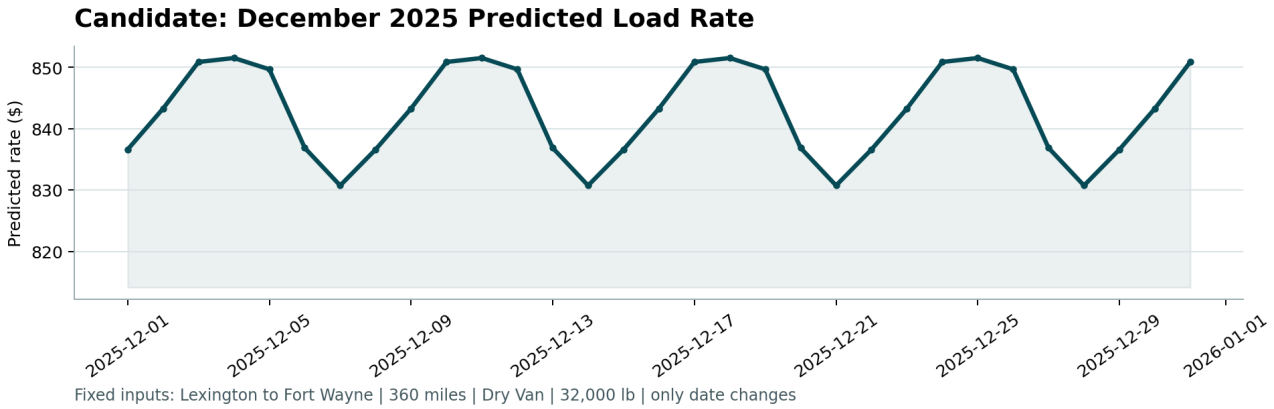


Figure 1. Candidate December 2025 predicted load rates for the fixed Lexington → Fort Wayne scenario.